MIL-F-11435A
18 MARCH 1954
SUPERSEDING
MIL-F-11435(TC)
10 SEPTEMBER 1951

MILITARY SPECIFICATION

FENDERS, MARINE, RUBBER-FILLED

This specification was approved by the Department of Defense for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope.—This specification covers cylindrical-shaped rubber-filled fenders to protect vessels and harbor craft from damage when going alongside another vessel or pier.

1.2 Classification.—Fenders shall be of the following types and sizes (see 6.1):

Type I —Canvas cover, rubber filled, fender.

Type II —Canvas cover, rubber filled, fender with outer rope cover.

Type I and type II completed fenders shall have the same external dimensions. Sizes shall be as follows:

Diameter (inches)		*Length (inches)*
4	by	18
7	by	24
9	by	30

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standard and publication of the issue in effect on date of invitation for bids form a part of this specification:

SPECIFICATIONS

FEDERAL

V-T-276 —Thread; Cotton.

CCC-T-191 —Textile Test Methods.

MILITARY

JAN-P-132 —Packaging and Packing for Overseas Shipment—Crates; Unsheathed, Wood; Nailed.

MIL-C-3133 —Cellular Rubber Products, General Purpose (Except Ebonite).

MIL-D-504 —Dyeing and Aftertreating Processes for Cotton Duck and Twill.

MIL-R-16060 (Ships)—Rope, Manila; and Substitutes.

MIL-T-3530 — Treatment, Mildew-Resistant, for Thread and Twine.

STANDARD

MILITARY

MIL-STD-129—Marking of Shipments.

(Copies of specifications and standard required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

PUBLICATION

CONSOLIDATED FREIGHT CLASSIFICATION COMMITTEE

Consolidated Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to the Consolidated Freight Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

270727—54—(415)

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3. REQUIREMENTS

3.1 General.—The fenders shall consist of a cylindrical synthetic rubber core having an endless manila rope vulcanized through the center with end loops projecting from each end of the core. The core shall be completely encased in a cotton tube double stitched at each end with machine or hand sewing. Type II fenders shall have a closely netted rope covering over the cotton tubing (see 3.2.4.2 and fig. 1).

3.2 Materials.—All materials used in the fabrication of the fenders shall be new and unused.

3.2.1 *Rubber.*—The cylindrical core shall be synthetic cellular rubber conforming to the requirements for RS12AB or RS43L of Specification MIL–C–3133 as specified (see 6.1).

3.2.2 *Cotton tubing.*—Cotton tubing shall be of the multiple ply type treated in conformance with the requirements of Class D Method 5c of Specification MIL–D–504. The warp of the cotton tubing shall have a minimum breaking strength of 700 pounds and the filling a minimum breaking strength of 200 pounds.

3.2.3 *Cotton thread and twine.* — Cotton thread used for machine sewing shall be 10 ply conforming to Type IIIA or IIIB of Specification V–T–276. An eight ply (commercial number 24) twine shall be used for hand sewing. Cotton thread and twine shall be treated in conformance to the requirements for Class I of Specification MIL–T–3530.

3.2.4 *Rope.* — Rope shall be three strand manila or sisal conforming to the requirements for Type M or S of Specification MIL–R–16060 (Ships).

3.2.4.1 *Center rope.* — An endless center rope shall be formed from one piece of manila rope, the ends joined with a short splice having three tucks each way. The splice shall be buried in the core and the resulting end loops shall project between 5 and 6 inches from the fender body. The rope size shall be in accordance with Table I.

TABLE I

Fender size (inches)	Diameter of rope (inches)	Circumference of rope (inches)
4 by 18	⅝	2
7 by 24	¾	2¼
9 by 30	1⅛	3½

3.2.4.2 *Rope covering.*—An idle rope and hitching rope of the same circumference (5/16-inch diameter) shall be used in the construction of the rope covering. Rope may be either manila or sisal. The ropes shall be tightly half-hitched. Each rope shall be not less than 30 feet long except where the covering is ended. When the hitching rope is within 2 feet of its total length, the ropes shall be interchanged and the idle rope employed as the half-hitching rope. A minimum of four half-hitches shall be used to secure the new idle rope to the former half-hitching rope. The covering shall be tightly half-hitched and shall result in a closely netted and tight covering.

3.3 Boat snap hooks. — When specified (see 6.1) the 4 by 18 inch fender shall have a commercial size number 2 galvanized boat snap hook on each of the end loops. The snap hooks shall be threaded on the center rope prior to splicing the rope and fabrication of the fender.

3.4 Workmanship.—Workmanship shall be of the highest quality throughout and in conformance with accepted commercial standard practice for this type of equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Lot size.—For purpose of inspection lot size shall be determined by the Government inspector.

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TABLE II

	Major defects AQL 2.5 percent defective	Minor defects AQL 6.5 percent defective
Dimensions	Diameter of center rope less than shown in Table I. Diameter of completed fender is ½ inch, or more, smaller than specified diameter.	Overall length of completed fender differs from specified dia. by ± 1 inch (see Table I). Projecting end loops exceed 6 inch or are less than 5 inches.
Center rope	Loose (not firmly attached to rubber core).	1 or more torn rope yarns. Frayed rope ends. Kinked rope.
Fender rope covering (Type II).	Loose (not tightly half hitched).	
Cotton tubing	Tears (any size). Open place (greater than 3-yarn diameter). Abrasion causing a weak place. Smash (any).	
Boat snap hooks (when specified).	Missing or improperly attached.	Incomplete coverage of galvanized coating.

4.2 SAMPLING

4.2.1 Sampling for visual and dimensional inspection of the completed fender shall be in accordance with Standard MIL–STD–105 for the specified Acceptable Quality Level.

4.2.2 *For test purposes.*—For each 1,000 fenders or less procured by the government, the contractor shall furnish the Government inspector a two foot length of cotton tubing representative of the tubing used in the fabrication of the fender. Each sample shall be marked to identify the lot represented by the sample.

4.3 **Inspection.**—The samples of completed fenders and samples of cotton tubing shall be inspected and tested for compliance with this specification.

4.3.1 *Classification of defects.*—For visual and dimensional inspection of the completed fenders, defects shall be classified in accordance with Table II.

4.3.2 *Strength test cotton tubing samples.*—If the cotton tubing sampled in accordance with 4.2.2 fails to meet the strength requirements specified (see 3.2.2), it shall be considered a defective, and the lot it represents shall be rejected.

4.4 TESTS

4.4.1 *Strength.* — The grab method test outlined in Method 5100 of Federal Specification CCC–T–191 shall be used to determine the strength of the cotton tubing material.

4.4.2 *Material tests.*—It is not mandatory for all materials, other than the cotton tubing, to be tested in conformance with the requirements of the referenced specifications in each individual case. However, the Government inspector will require these referenced tests whenever, in his judgment, they are necessary to ascertain that the quality of the materials used conforms to the requirements of the referenced specifications.

5. PREPARATION FOR DELIVERY

5.1 **For domestic shipment—immediate use.**—Fenders shall be packed in conform-

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ance with applicable Consolidated Freight Classification Rules and in a manner acceptable to common or other carrier at the lowest applicable rate.

5.2 For domestic shipment and storage, or overseas shipment.—Fenders of one type and size shall be packed in unsheathed crates conforming to Style A or C of Specification JAN-P-132. The fenders shall be secured and shrouded, and the crates reinforced with steel straps as described in the appendix to Specification JAN-P-132.

5.3 Marking.—In addition to any special marking required by procurement documents, contents and shipping containers shall be marked in conformance with Standard MIL-STD-129. The nomenclature shall be "Fenders, Marine, Rubber-Filled".

6. NOTES

6.1 Ordering data.—Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type, size, and number of fenders required.

(c) Type of rubber core (Transportation Corps is limited to use of Type RS12AB).

(d) If fenders are to be equipped with boat snap hooks.

(e) Whether preparation for domestic shipment—immediate use, domestic shipment and storage, or overseas shipment is required (see 5.1 and 5.2).

(f) Whether special marking is required (see 5.3).

Patent Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Transportation Corps
Navy—Bureau of Ships
Air Force

Other Interest:
Army—EO.

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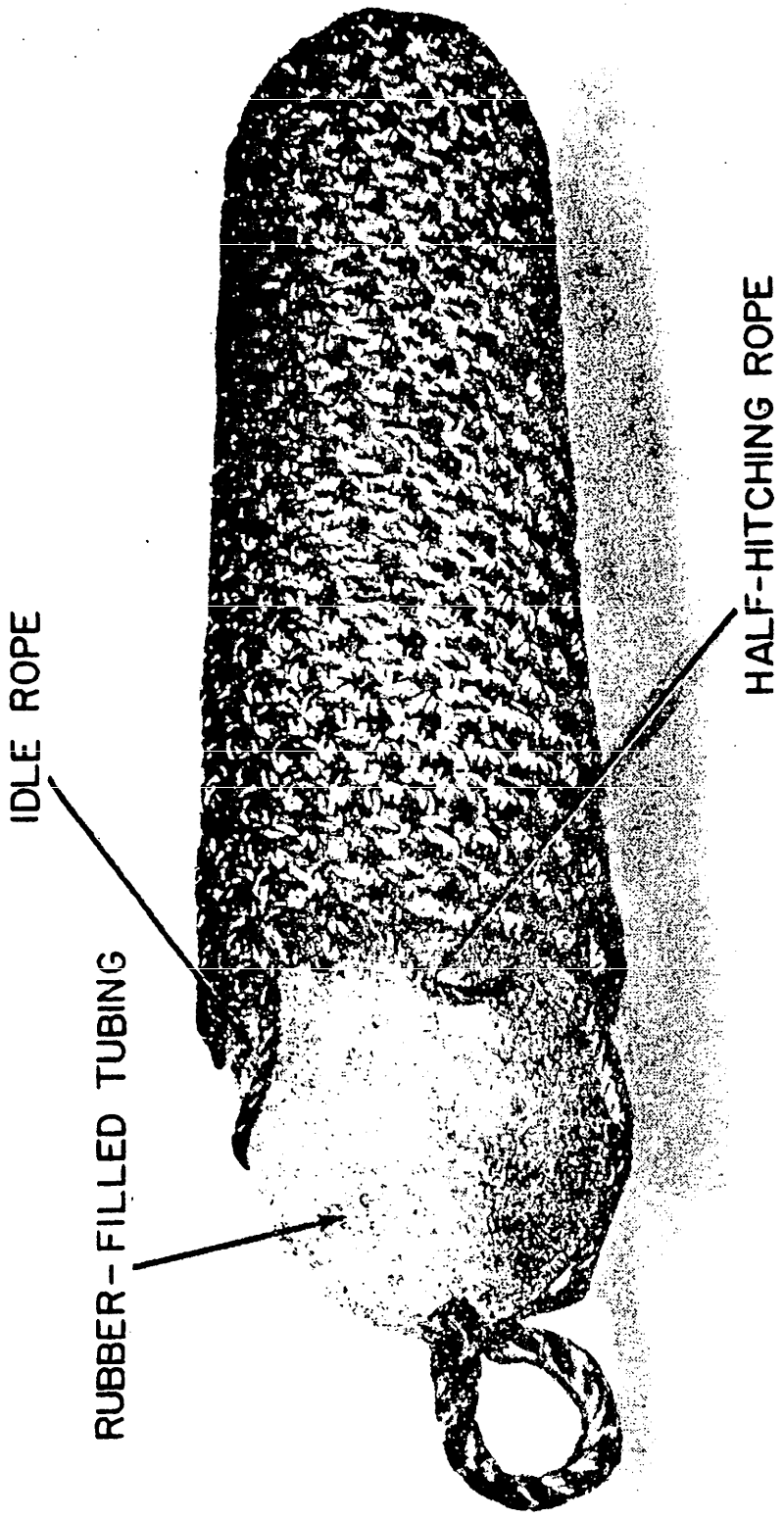


FIGURE 1.—*Fender, Marine, rubber-filled.*

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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